

Clovis SFEIR

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Profile

IMT Atlantique engineering student specialising in quantitative systems and trading infrastructure. Built five systems (C++20/Rust/Python) benchmarked against industry references or labeled ground-truth: an Ornstein-Uhlenbeck crude-spread trading engine, an HFT limit-order-book matching engine at **sub-microsecond P50 latency**, a finite-difference reservoir PDE solver verified to **2nd-order convergence**, and an IEC-104 ICS intrusion detector with **100% attack recall**.

Education

IMT Atlantique – Engineering Degree in Software Engineering (Apprenticeship) *Sep 2025 – Jun 2028*
Nantes, France *Algorithms & Data Structures, OS, Computer Networks, Distributed Systems, Cloud Computing, AI*
Bachelor in Computer Science *Sep 2022 – Jun 2025*
La Rochelle, France *Class rank: 1/85 | Software Engineering, Databases, Cybersecurity*
HFT Macro Placement Challenge 2026 (Hudson River Trading/Partcl) — **24th/124** (top 20%); hybrid analytical placer: force-directed global placement with timing-aware legalization; proxy cost **1.095** beats SA baseline (2.125) and RePlace (1.458); field includes Stanford, Princeton, ETH Zürich.

Projects

Crude Oil Spread Trading Engine

C++20, stochastic processes, order book

- Models refining margins (3:2:1 crack spread, generalized N:M:K) and mean-reverting inter-crude spreads via an Ornstein-Uhlenbeck process (Euler-Maruyama) with a rolling z-score signal, matched by a price-time-priority order book.
- OU stationary distribution verified against theory (500K-step sim); crack-spread formula checked to 10^{-9} ; order-book submit at **720 ns** (non-crossing)/**1.24 μ s** (crossing); 58 assertions/21 tests, ASan+UBSan clean.

HFT Limit Order Book Matching Engine

C++20, cache-aware layout, intrusive data structures

- Price-time-priority matching engine (limit/market/cancel) with intrusive doubly-linked price-level lists (**O(1) cancel** via pointer unlink) and pre-allocated object pooling to keep the hot loop free of `malloc/new`.
- Benchmark suite across three synthetic order-flow regimes: **2.3M+ msg/s** single-threaded throughput, **P50 0.6–1.0 μ s** / P99 2–4 μ s latency, **zero hot-path allocations**; unit tested.

Reservoir Diffusivity PDE Solver

C++20, OpenMP, finite-difference

- Explicit FTCS and implicit Crank-Nicolson finite-difference schemes for the reservoir diffusivity (pressure-transient) equation; verified via grid-refinement convergence study against the closed-form erfc analytical solution – error ratio converges to **4.0** as dx halves (textbook 2nd-order accuracy) across 5 refinements.
- OpenMP-parallelized stencil/assembly loops; measured (not assumed) scaling identifies the Amdahl's-law-bounded sequential Thomas solve and memory-bandwidth-bound behavior below L3 size; 13 test cases, 40,300 assertions.

Engineering Docs RAG

Python, BM25, information retrieval

- From-scratch Okapi BM25 retrieval engine (chunking + ranking) for engineering documents (datasheets, HAZOP-style procedures); pluggable extractive/LLM generator interface, zero paid API calls in tests/CI/demo by design.
- Hand-derived BM25 scoring verified to 9 significant figures; measured **precision@1 0.90**, recall@5 1.00, **MRR 0.944** on a 40-question labeled set; ~3,330 docs/s ingest; 53 tests, zero runtime dependencies.

ICS Intrusion Detection (IEC 60870-5-104)

C++20, protocol parsing, explainable rules

- Rule-based detector for IEC 60870-5-104 (no auth/encryption by protocol design) – the actual telecontrol protocol behind the 2016 Kyiv blackout – mapped to real attack patterns: control commands to unexpected breaker objects (the exact method used), sequence-number gaps (replay/desync), u-format/command floods.
- Parses APCI+ASDU at **39.5 ns (25.3 M msg/s)**; end-to-end parse+detect 337 ns; **100% recall, 0 false positives** on a labeled synthetic trace (100 records, 15 attacks across 5 rules); 153 assertions/37 tests, ASan+UBSan clean.

Work Experience

Infotel

AI Engineer Apprentice – Lab IA

Nantes, France

Sep 2025 – Jun 2028

- Built a Rust-based autonomous software-delivery harness that takes a complete specification through architecture synthesis (subsystems, interface contracts), epic/slice decomposition, and TDD-first generation to a tested, deployable application end-to-end; automated compile+test quality gates with iterative self-repair, independent code-review pass, and cross-slice conflict detection before delivery; 928 Rust tests; validated end-to-end producing a fully working, 100%-test-passing application directly from a single specification.

Excelia Group

Software Developer Intern / AI Software Engineer Intern

La Rochelle, France

Two internships: Jan–Mar 2025 & May–Jul 2024

- RAG pipeline (chunking, OpenAI embeddings, flat FAISS, cross-encoder re-ranking): **40% reduction** in lookup time vs. keyword-search; cut p95 latency 30–40% via embedding cache + asyncio parallelisation; Prometheus/Grafana, Docker, GCP.
- Real-time interview simulator (Next.js/React, GPT-4o, ElevenLabs TTS): streaming responses, scenario-branching state machine, A/B prompt-scoring harness.

Technical Skills

ML / Quant

Ornstein-Uhlenbeck, finite-difference PDE, BM25/information retrieval, EWMA/CUSUM change-detection, Kalman filter, LightGBM/XGB/CatBoost

Perf / Systems

OpenMP, cache-aware layout, acquire/release atomics, intrusive linked lists, ICS/SCADA protocol parsing (Modbus, IEC-104)

Languages

C++20, Python, Rust, TypeScript, SQL

Infrastructure

Docker, tokio async, Prometheus/Grafana, GitHub Actions CI, GCP, Redis, ZeroMQ

Frameworks

Flask, Node.js/Express, React/Next.js, REST/gRPC